

Complete the exercise and write the learning in one of the cell

1. Problem1: A fair coin is tossed 100 times. Calculate:

- Number of heads
- Number of tails
- Experimental probability of Head
- Experimental probability of Tail

2. Problem 2: check Effect of the number of trials

Repeat the coin-toss experiment for: $n=10, 100, 1,000, 10,000$

3. Rolling a die: A fair die is rolled 6,000 times.

Calculate the experimental probability of each outcome: $S=\{1,2,3,4,5,6\}$

4. Addition rule for disjoint events

A die is rolled 10,000 times.

Define: A =getting 1, B = getting 6, check the events are joint or disjoint

Verify: $P(A \cup B) = P(A) + P(B)$

5. Addition rule for overlapping events

For one die roll, define: A =getting an even number $A=\{2,4,6\}$ and B =getting a number greater than 3 $B=\{4,5,6\}$

Verify: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

6. Multiplication rule for independent events

A coin is tossed and a die is rolled 10,000 times.

Define:

- obtaining Head
- obtaining an even number

Verify: $P(A \cap B) = P(A)P(B)$